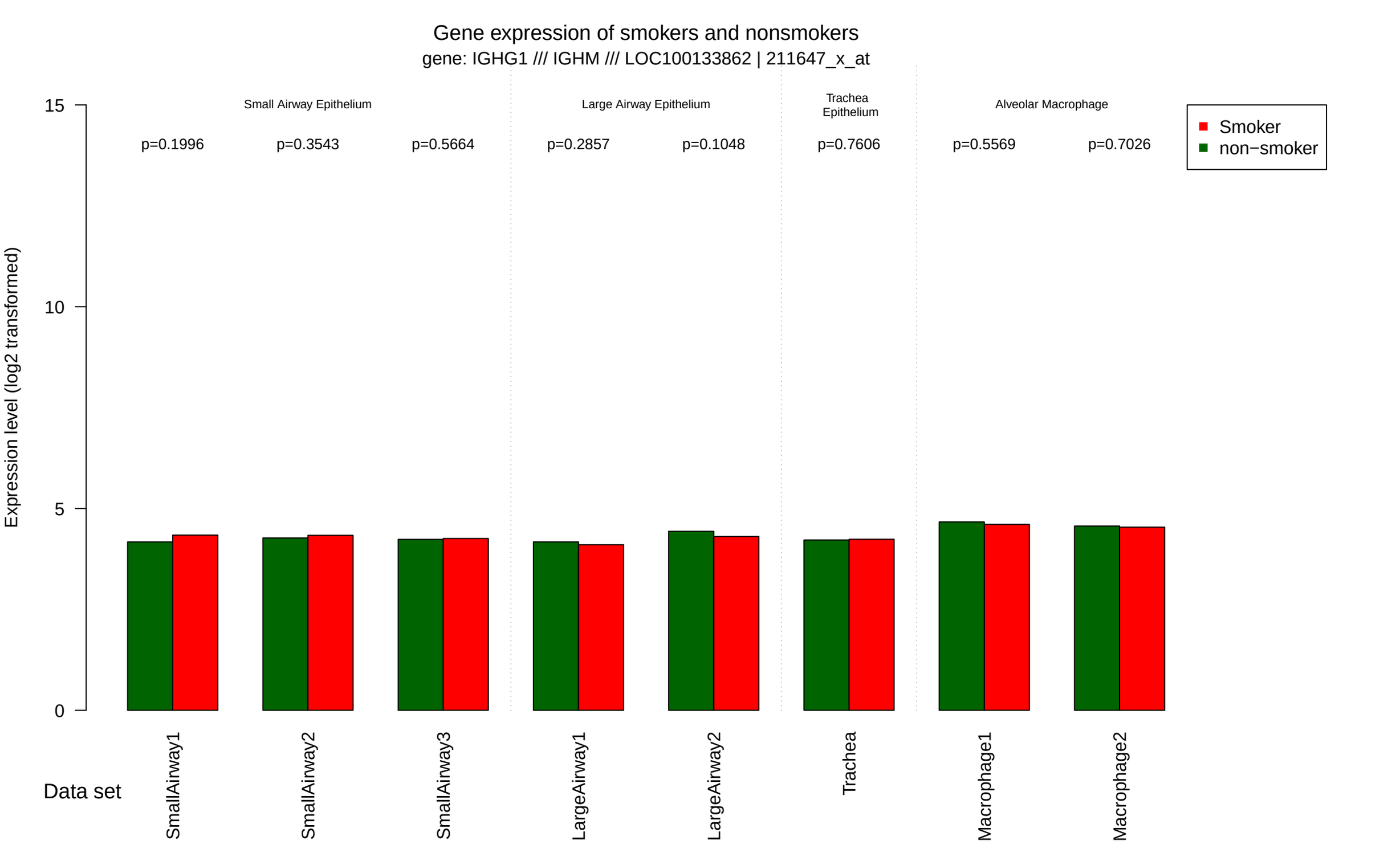
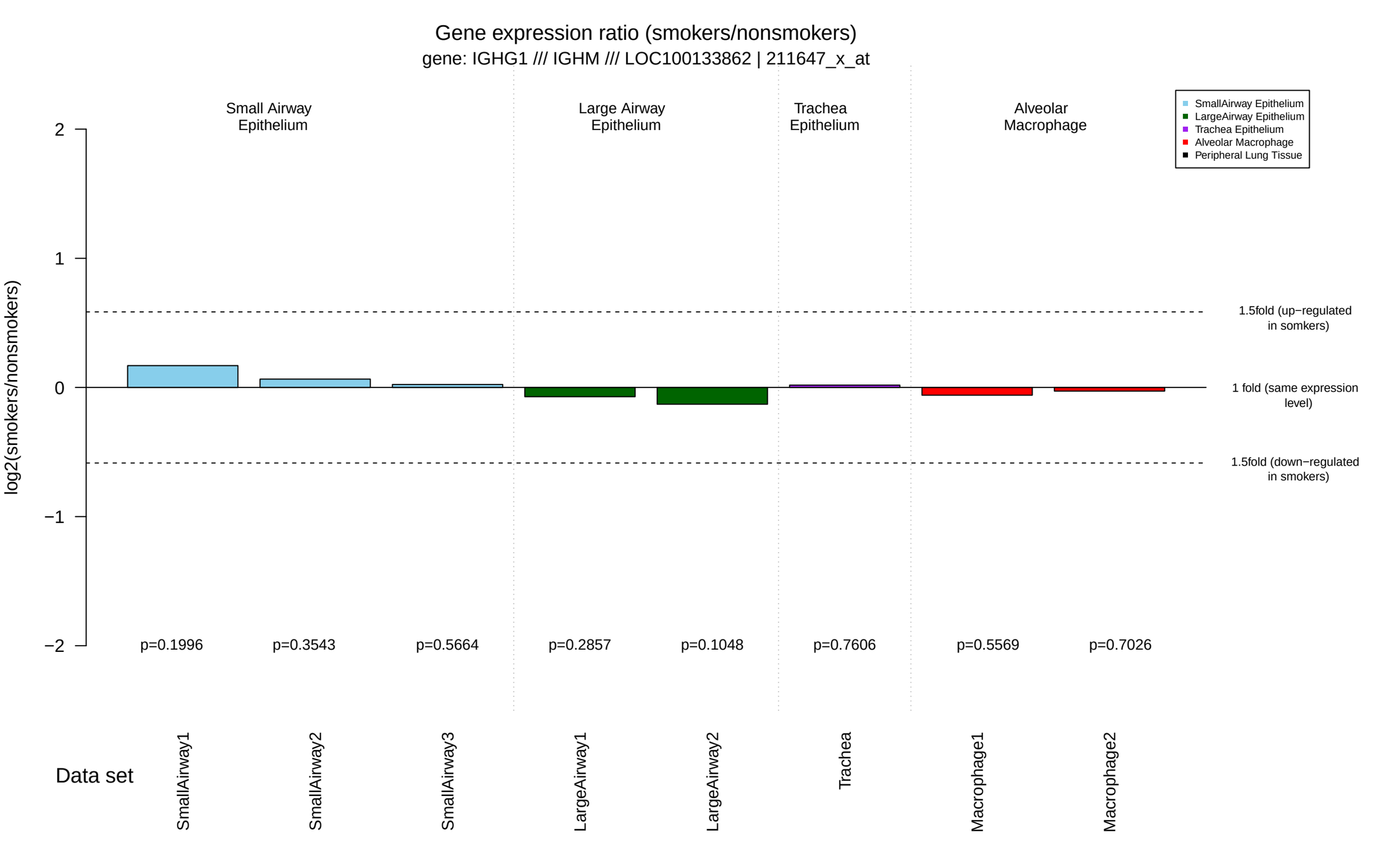
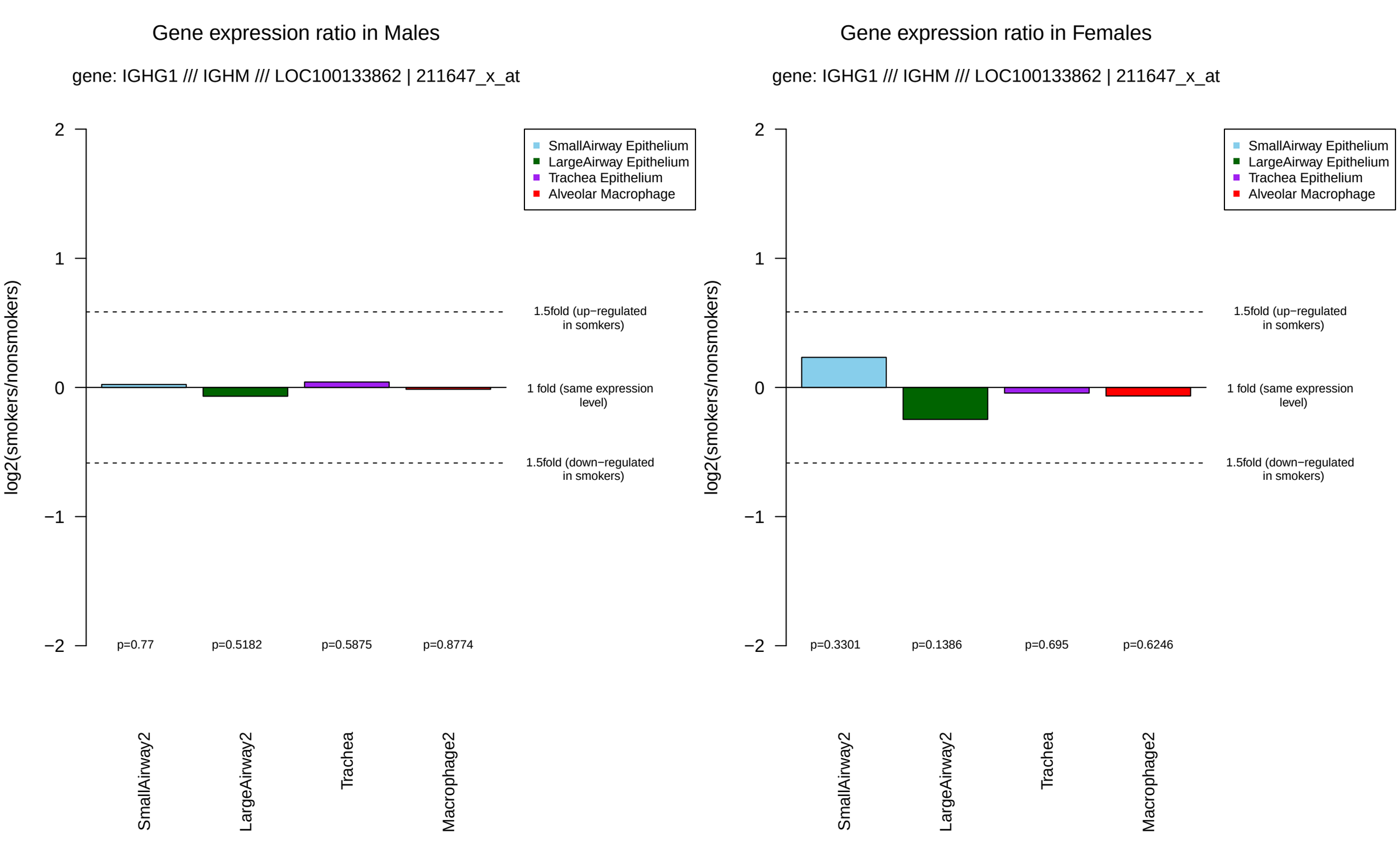
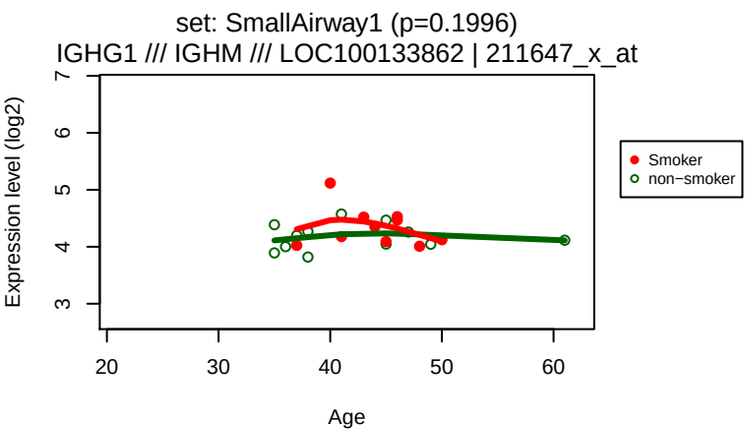
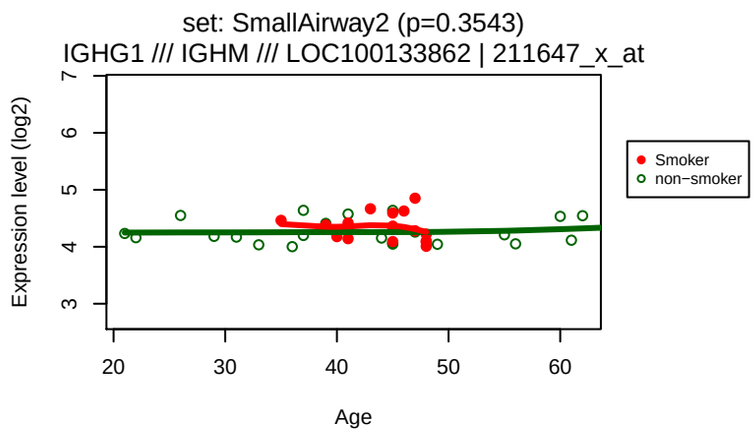
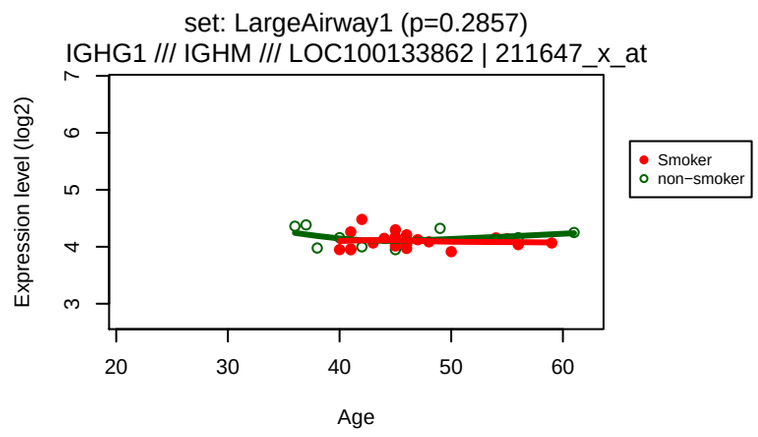

Gene expression of smokers and nonsmokers
gene: IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1996
p=0.3543
p=0.5664
p=0.2857
p=0.1048
p=0.7606
p=0.5569
p=0.7026
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

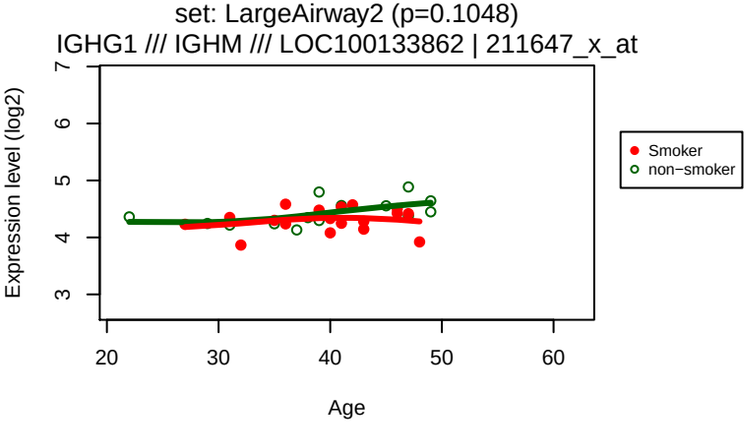

Gene expression ratio (smokers/nonsmokers)
gene: IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1996
p=0.3543
p=0.5664
p=0.2857
p=0.1048
p=0.7606
p=0.5569
p=0.7026
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

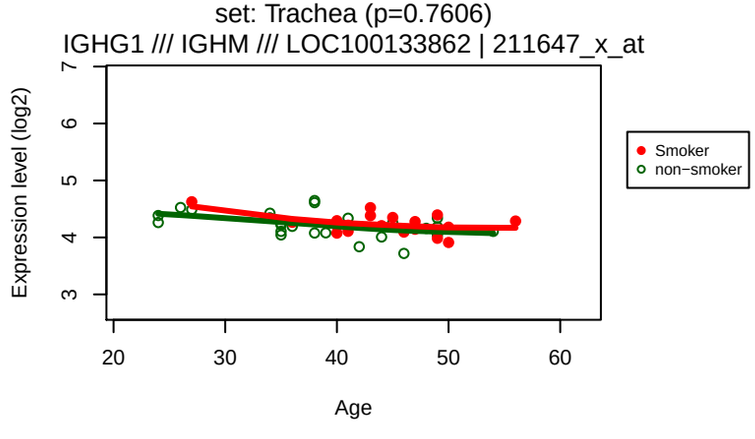

Gene expression ratio in Males
gene: IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.77
p=0.5182
p=0.5875
p=0.8774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3301
p=0.1386
p=0.695
p=0.6246
SmallAirway2
LargeAirway2
Trachea
Macrophage2

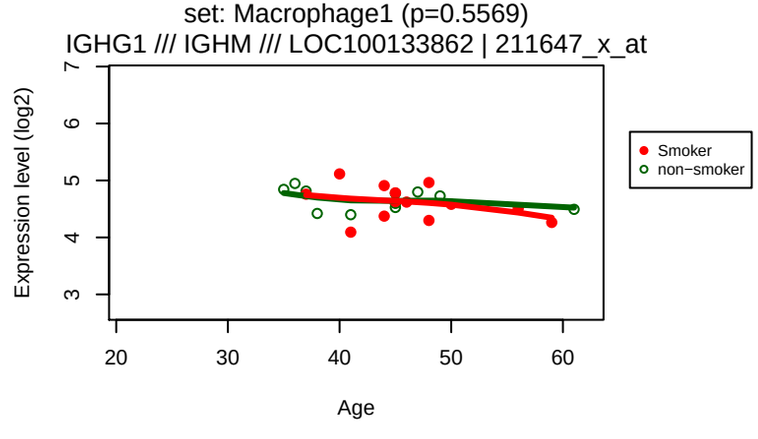

set: SmallAirway1 (p=0.1996)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker

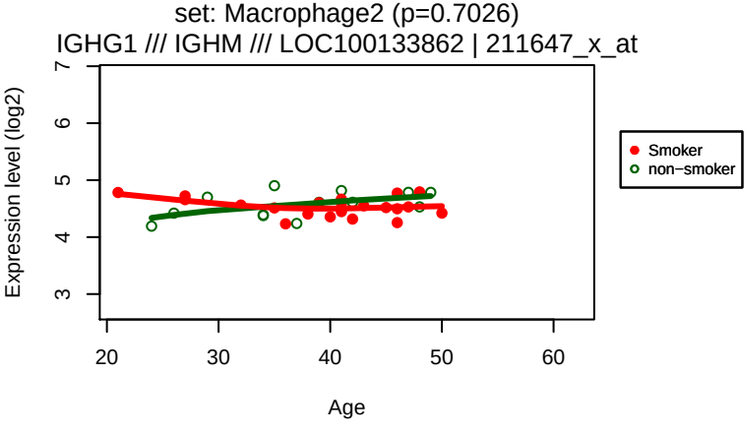

set: SmallAirway2 (p=0.3543)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker

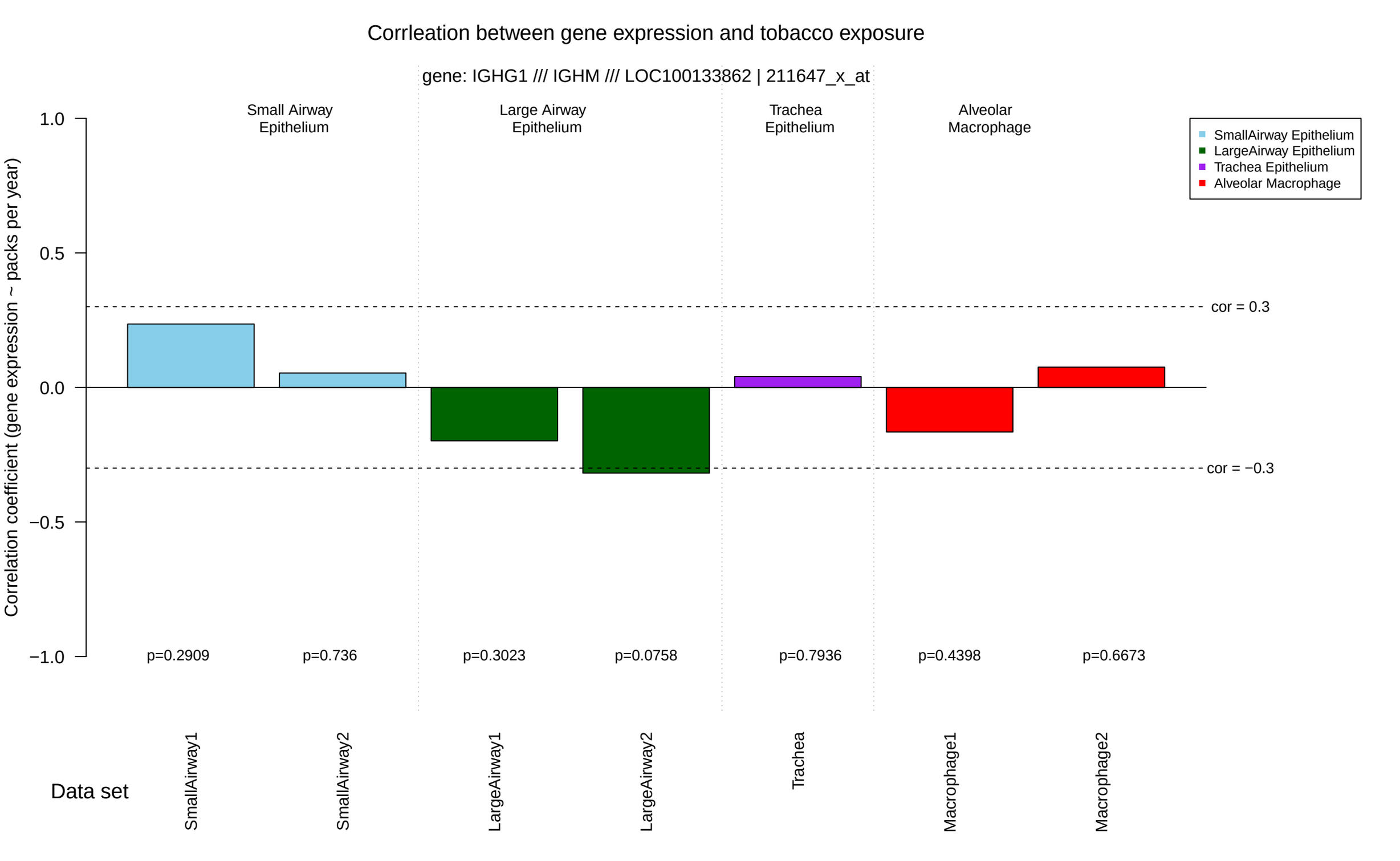

set: LargeAirway1 (p=0.2857)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1048)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7606)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5569)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7026)
IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHG1 /// IGHM /// LOC100133862 | 211647_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2909
p=0.736
p=0.3023
p=0.0758
p=0.7936
p=0.4398
p=0.6673
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2